

SOUTHAMPTON STATISTICS SUMMER SCHOOL

Introduction to Machine Learning

Brody Hannan

University of Southampton · 30 July 2026

brody.hannan@soton.ac.uk | github.com/BHannan01/introduction-to-machine-learning



Today's Workshop

PART 1 · THEORY

45 minutes

1. Why ML for social science?
2. What ML is — and what it is not
3. The ML workflow
4. Supervised vs. unsupervised learning
5. Decision trees
6. Random forests
7. Gradient boosting
8. Model evaluation
9. Ethical limits

PART 2 · PRACTICAL

75 minutes

1. Google Colab notebook (Python)
2. 1994 US Census income data
3. Build and compare three models
4. Interpret variable importance
5. Reflection questions throughout
6. Extension: clustering

github.com/BHannan01/introduction-to-machine-learning

SECTION 01

Why Machine Learning?

Activity: Who Earns More Than \$50K?

Person	Age	Education	Occupation	Hrs/week
Alex	28	High school	Service worker	35
Jordan	45	Bachelor's	Manager	55
Sam	32	Master's	Tech worker	40
Casey	52	No diploma	Farming	45
Morgan	38	Bachelor's	Self-employed	60

Take 60 seconds. Rank these five people from most to least likely to earn >\$50K per year.

Then ask yourself:

- What information did you use?
- What would have made you more confident?
- Can you write down the exact rule you applied?

THIS IS EXACTLY WHAT ML DOES

A machine learning model learns those rules automatically from data — for 48,842 real people at once — and tells you which rules it used.

But what if the data was much larger?

Chetty et al. (2018)

- ML applied to US tax records for **20 million families**
- Identified neighbourhood-level predictors of upward mobility
- Social capital, school quality and commute times predicted adult income better than individual-level variables
- Produced a policy-relevant map of opportunity at a scale regression alone could not reach

Your intuitive ranking of five people is slow, inconsistent and hard to audit. ML makes the process **systematic, scalable and transparent** — but introduces its own risks.

Chetty, R., et al. (2018). The opportunity atlas. NBER Working Paper 25147.

SECTION 02

What Is Machine Learning?

"AI" Is Not the Same as Machine Learning

Artificial Intelligence

rule-based & expert systems, game playing

Machine Learning

decision trees, random forests, SVMs, clustering · since the 1950s

Deep Learning

multi-layer neural networks · images, speech, text · prominent from 2012

Large Language Models

GPT-4, Claude, Gemini, Llama · public from 2022

A brief history

- **1950s–70s** ML emerges within AI research
- **1980s** ML and AI diverge
- **1990s** ML formalised as its own discipline
- **2010s** Deep learning revives neural networks at scale
- **2020s** LLMs make "AI" visible to the public

Machine Learning is not...

- the same as AI
- just chatbots
- new

ML vs. Traditional Statistics

	Traditional Statistics	Machine Learning
Primary goal	Explain relationships	Predict outcomes accurately
Starting point	Theory and hypothesis	Patterns in data (inductive)
Model form	Pre-specified	Learned from data
Output	Coefficients, p-values, CIs	Predictions, accuracy, importance
Core strength	Inference and interpretability	Flexibility and performance
Core limitation	Rigid assumptions	Less directly interpretable

They are complementary tools, not competitors.

Supervised vs. Unsupervised Learning

Supervised learning

- Has a labelled outcome
- Classification (categorical) or regression (continuous)
- Methods: decision trees, random forests, gradient boosting

EXAMPLE

Given census records with known incomes, predict whether a new person earns >\$50K.

Unsupervised learning

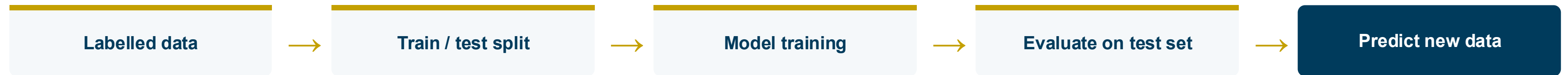
- No outcome variable
- Clustering — discover sociodemographic types
- Dimensionality reduction — PCA for survey scales
- Extension activity in the practical

EXAMPLE

Given survey responses, group respondents into typologies of user.

Today focuses on supervised classification, with a clustering extension at the end.

The Core Idea: Learning from Labelled Data



Training set · 80%

Test · 20%

Training set — 80%

The examples the model studies. It sees the features (age, education, hours...) *and* the correct answer — did each person earn >\$50K? — and learns the patterns that link them.

Test set — 20%

A separate group of people the model never sees while learning. We reveal their answers only afterwards, to check whether it truly learned the patterns — or just memorised the training rows.

We choose the split. 80/20 is common, but 70/30 or 90/10 also work — more training data usually helps, but too small a test set gives a noisy score. The two sets never overlap: no person appears in both, so the test set is genuinely unseen.

Baseline Model Accuracy

Our dataset: 75% earn $\leq \$50K$, 25% earn $> \$50K$

A model that predicts " $\leq \$50K$ " for everyone gets:

Accuracy

share of all predictions that are correct ·
acceptable $\approx 80\%$ · good $\approx 85\%$ · excellent $\approx 88\%+$

Recall ($> \$50K$)

share of true $> \$50K$ earners the model actually catches — where accuracy rewards getting the easy majority right, recall counts only the hard $> \$50K$ cases · **acceptable ≈ 0.55 · good ≈ 0.65 · excellent ≈ 0.75**

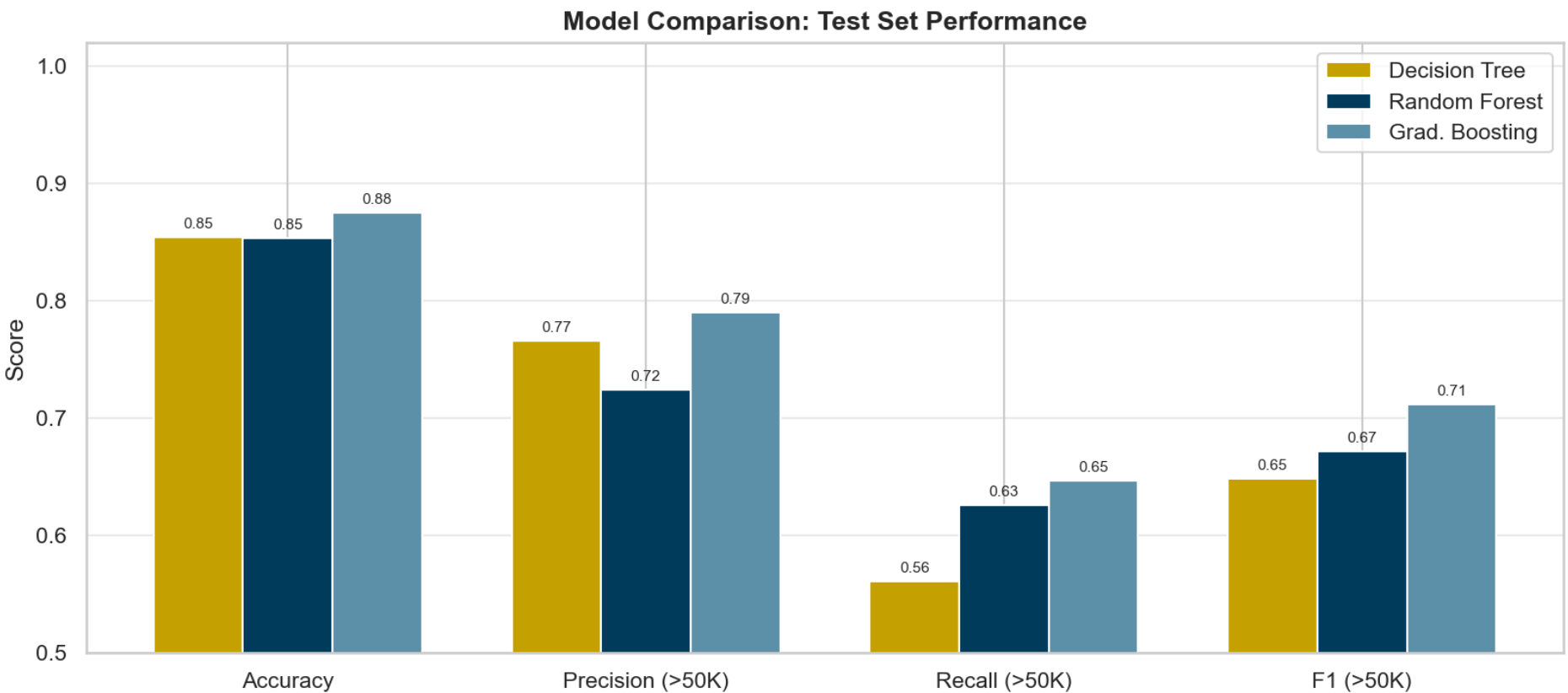
F1 ($> \$50K$)

$F1 = 2 \cdot (P \times R) / (P + R)$; with recall = 0 here, F1 collapses to 0 ·
acceptable ≈ 0.60 · good ≈ 0.68 · excellent ≈ 0.75

What to report:

- Baseline accuracy
- Precision for the minority class
- Recall for the minority class
- F1 score

Our model must beat **75% accuracy** AND show meaningful recall for the minority class to be useful.



SECTION 03

Decision Trees

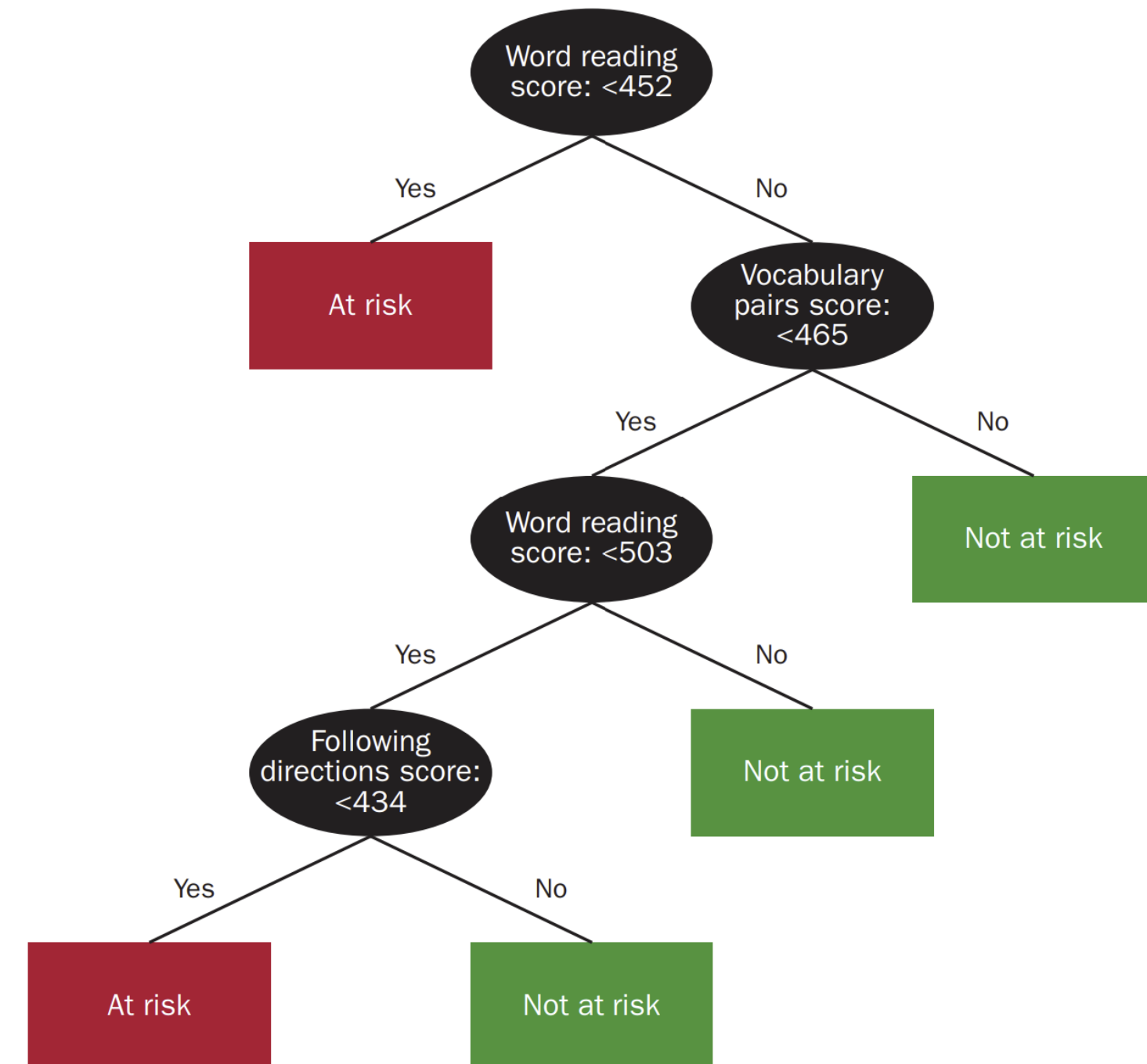
Decision Trees: How They Work

- A sequence of binary yes/no questions (splits) that classifies each observation
- At each node, the algorithm chooses the feature and threshold that best separates the classes
- Splitting criterion: **Gini impurity** — how mixed a node's classes are (0 = pure; 0.5 = a 50/50 mix). At each node the tree picks the split that most reduces impurity, driving each group toward a single class.
- Grows until leaves are pure or a stopping rule is reached — e.g. a maximum depth, a minimum number of observations per leaf, or too small a gain in purity

Strengths

Intuitive and visual · handles mixed data types · requires minimal preprocessing

Figure 2. Grade 1 classification rules under the classification and regression tree model



Koon, Petscher & Foorman (2014), REL Southeast. Reproduced from Hall (2025).

The Problem with Single Trees

High variance

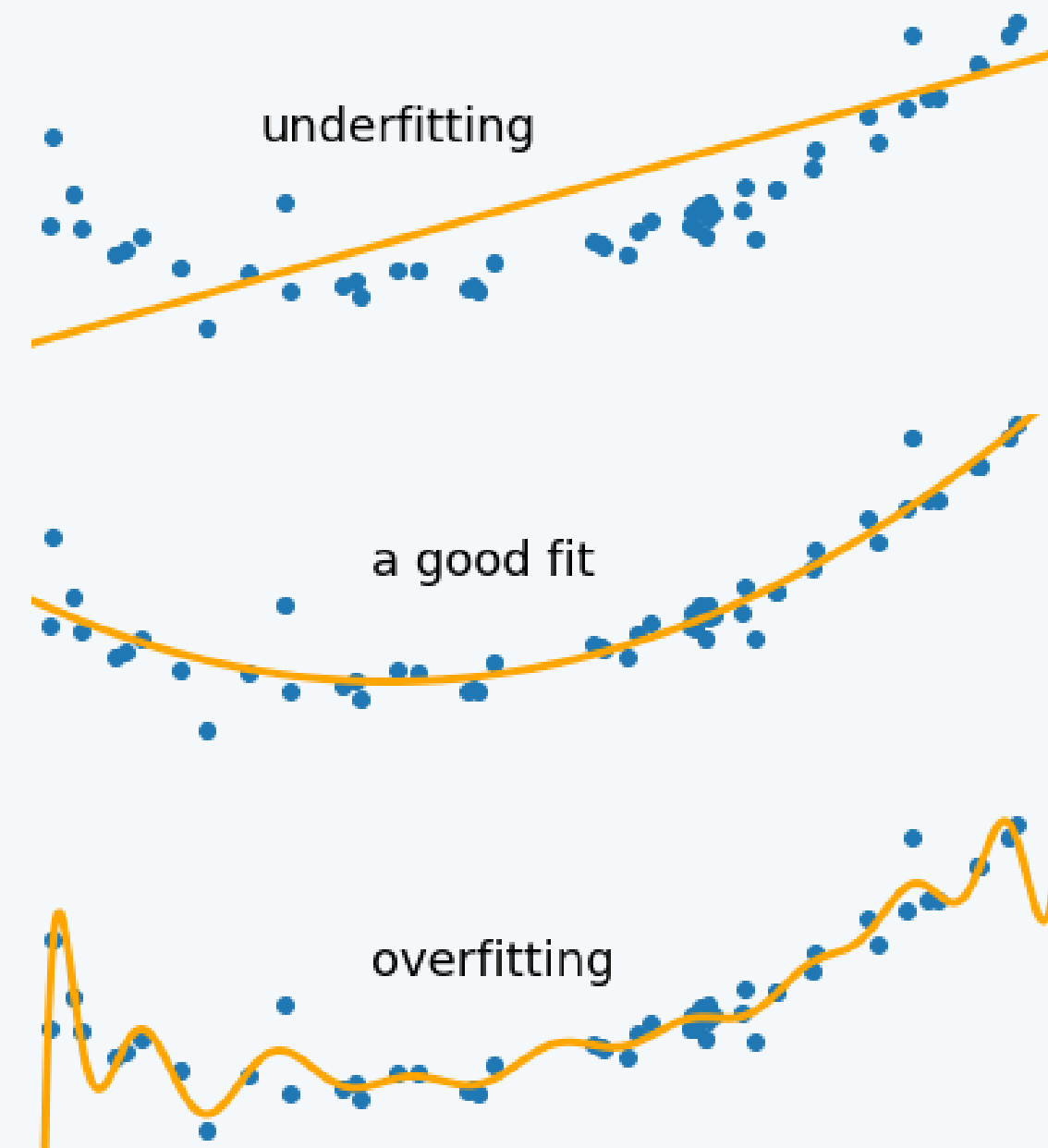
A small change in the training data produces a completely different tree.

Overfitting

A deep tree memorises the training data and fails on new cases.

Asking one person a yes/no question is unreliable.
Asking **500 people** and taking a majority vote is much more robust.

Underfitting vs. overfitting



SECTION 04

Combining Many Trees

Ensemble methods — many models, one stronger prediction

Random Forests: Wisdom of the Crowd

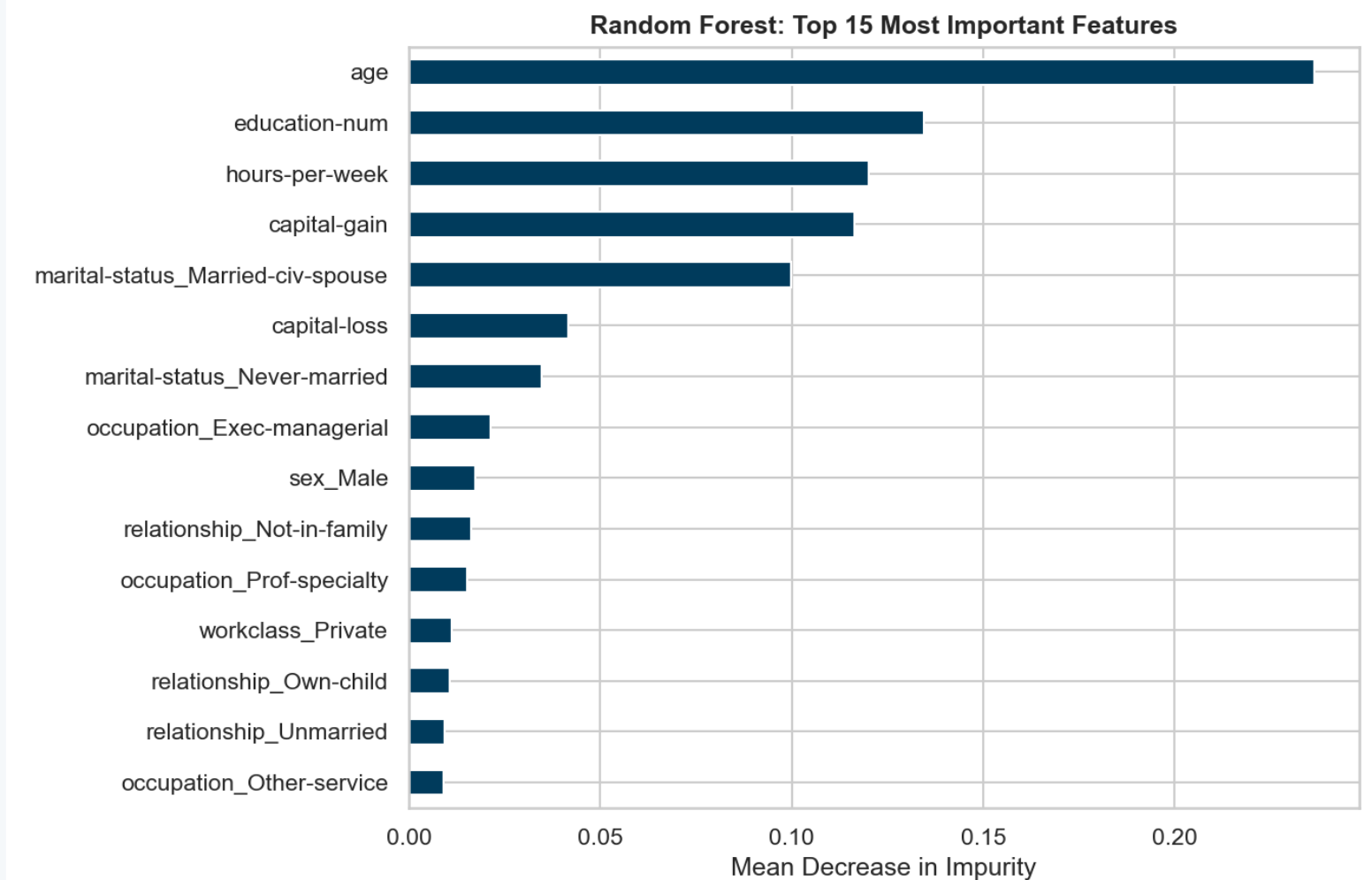
1. Draw a bootstrap sample (random rows, with replacement)
2. Grow a full tree, but at each split consider only a random subset of features
3. Repeat for N trees (100–500)
4. Final prediction: majority vote (classification) or average (regression)

Why it works

Each tree sees different data and features; errors are uncorrelated; averaging cancels them out.

Variable importance (MDI)

The reduction in Gini impurity summed over all splits — what the model leaned on most.

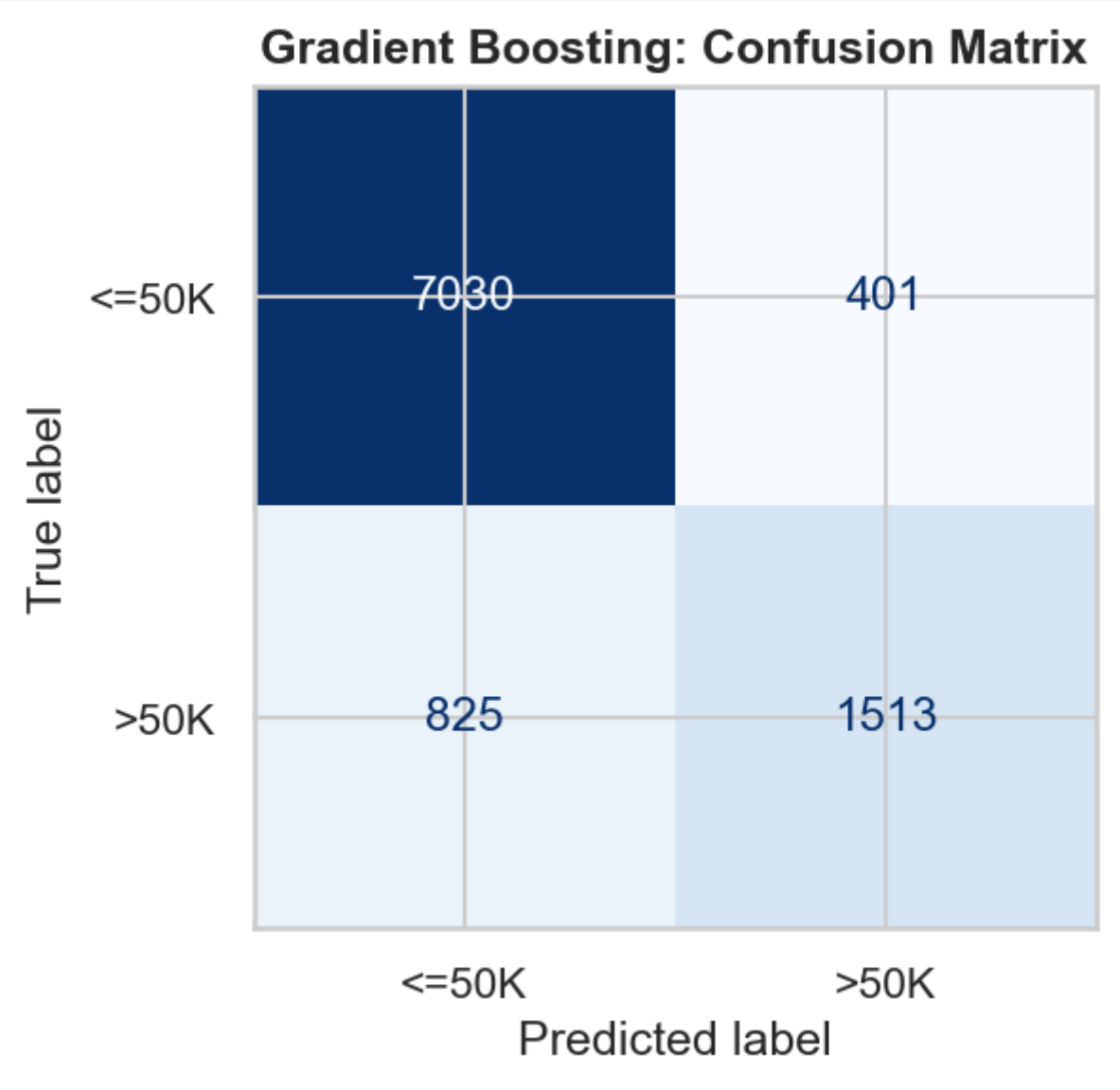


Gradient Boosting: Learning from Mistakes

- 1. Start with a simple initial prediction
- 2. Fit a shallow tree to the *residuals* (errors) of the current model
- 3. Add that tree, scaled by a learning rate
- 4. Repeat for many iterations

RF grows trees in parallel (independent). **GB** grows them sequentially — each corrects the previous.

Parameter	Effect
n_estimators	Number of trees
learning_rate	Step size per tree
max_depth	Tree complexity
subsample	Row sampling fraction



Gradient boosting reaches **~87% accuracy** — clearing the 75% baseline and catching a real share of the >\$50K minority.

Comparing the Three Methods

	Decision Tree	Random Forest	Gradient Boosting
Training speed	Fast	Moderate	Slow
Predictive accuracy	Moderate	High	Very high
Interpretability	High	Low	Low
Overfitting risk	High	Low	Moderate
Tuning required	Minimal	Some	Significant

Tuning required = how much you must hand-set the model's settings — tree depth, number of trees, learning rate — to get good results.

Overfitting risk rises with flexibility: a lone deep tree memorises noise; a forest averages many de-correlated trees to cancel it; boosting re-fits errors, so it creeps back if run too long.

Practical advice: start with a random forest — robust, minimal tuning, free variable importance. Move to gradient boosting if you need the last increment of accuracy.

SECTION 05

Model Evaluation

Evaluating Performance: The Confusion Matrix

	Predicted >\$50K	Predicted ≤\$50K
Actual >\$50K	TP ✓ True Positive	FN ✗ False Negative
Actual ≤\$50K	FP ✗ False Positive	TN ✓ True Negative

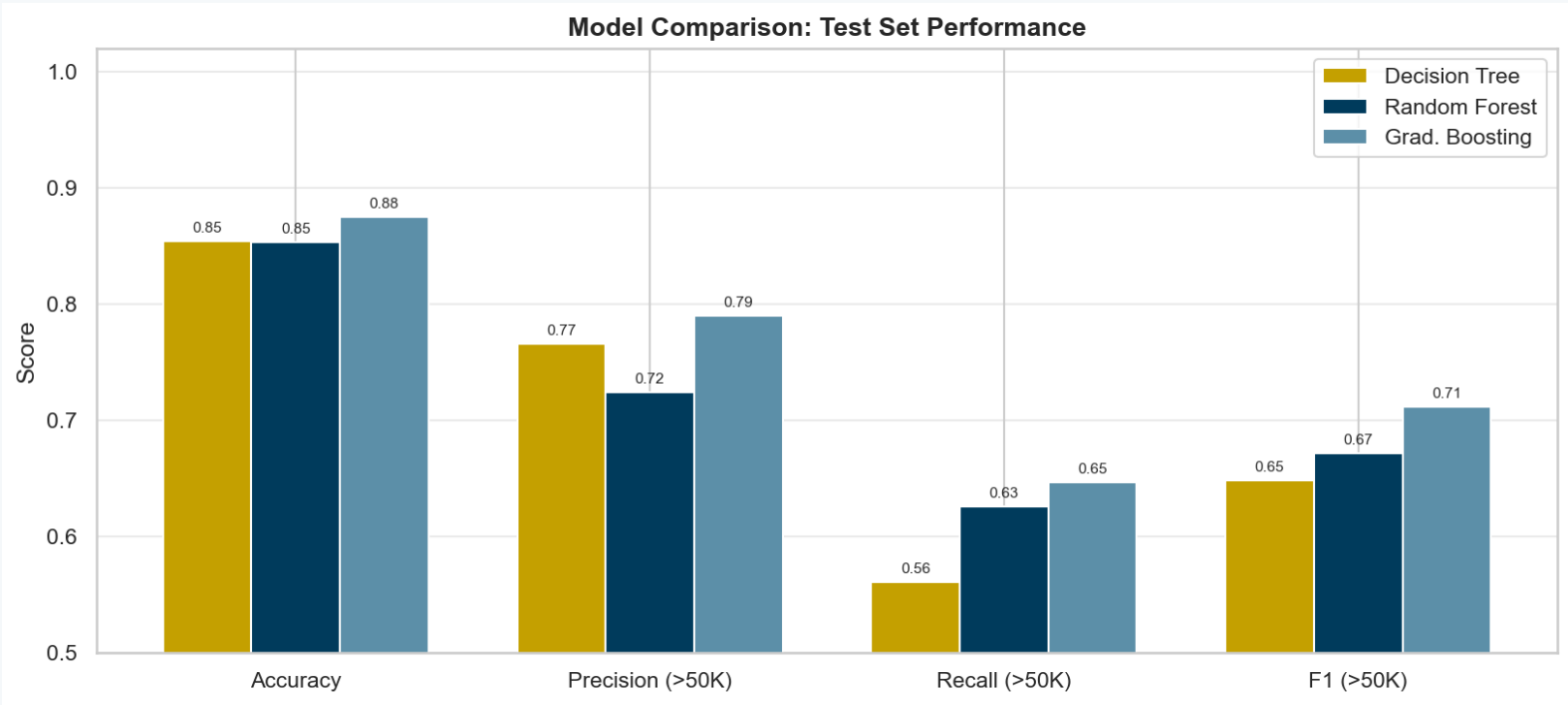
Accuracy — share of all predictions correct: $(TP + TN) / \text{all}$

Precision — of those flagged >\$50K, the share that truly were: $TP / (TP + FP)$

Recall — of all true >\$50K earners, the share we caught (predicted correctly): $TP / (TP + FN)$

F1 — one score balancing precision and recall: $2 \cdot (P \times R) / (P + R)$

- "Predict ≤\$50K always" = **75% accuracy** but **0% recall** for >\$50K
- For this dataset, published benchmarks sit around **85–87% accuracy** (Kohavi, 1996) — a realistic ceiling for social-science prediction of this kind



SECTION 06

Unsupervised Learning: Clustering

Finding Hidden Groups in Your Data

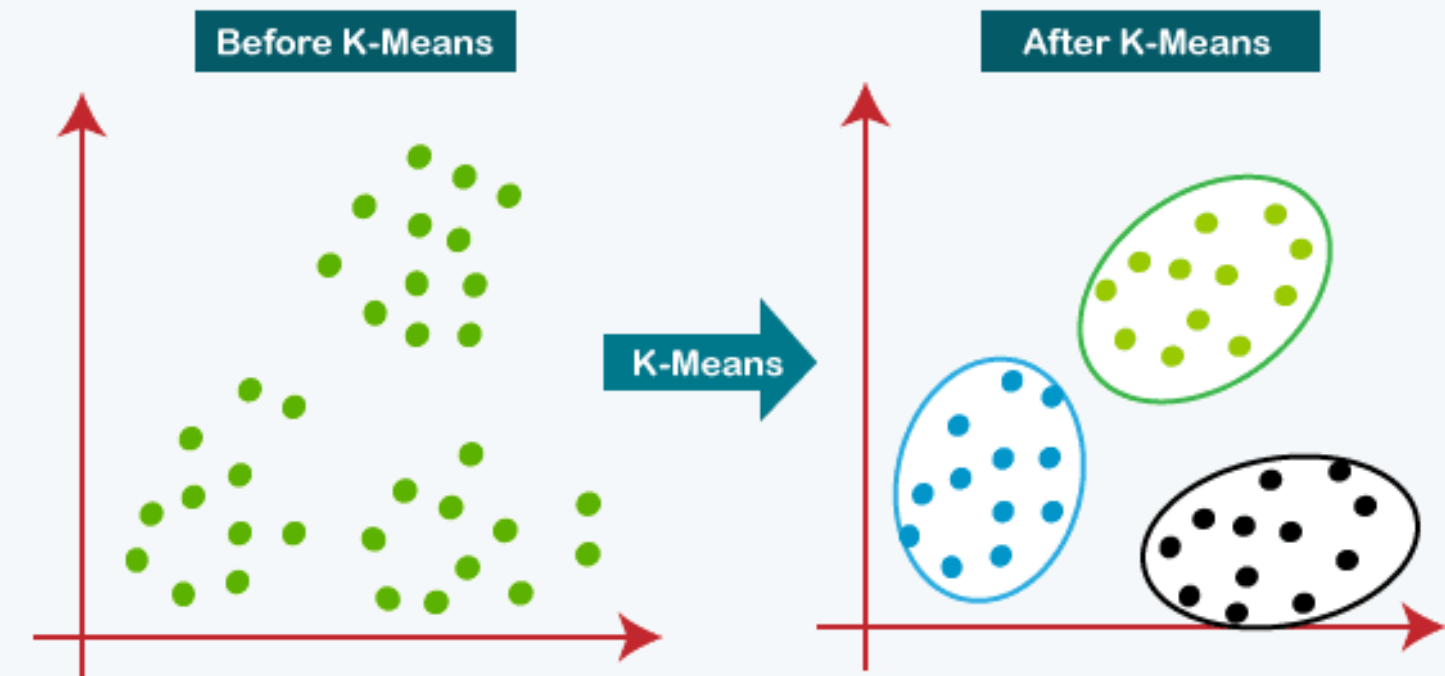
- No outcome variable — find natural groupings
- Groups similar observations; dissimilar from other groups
- Output: a cluster label for each observation, plus a characterisation

K-means algorithm

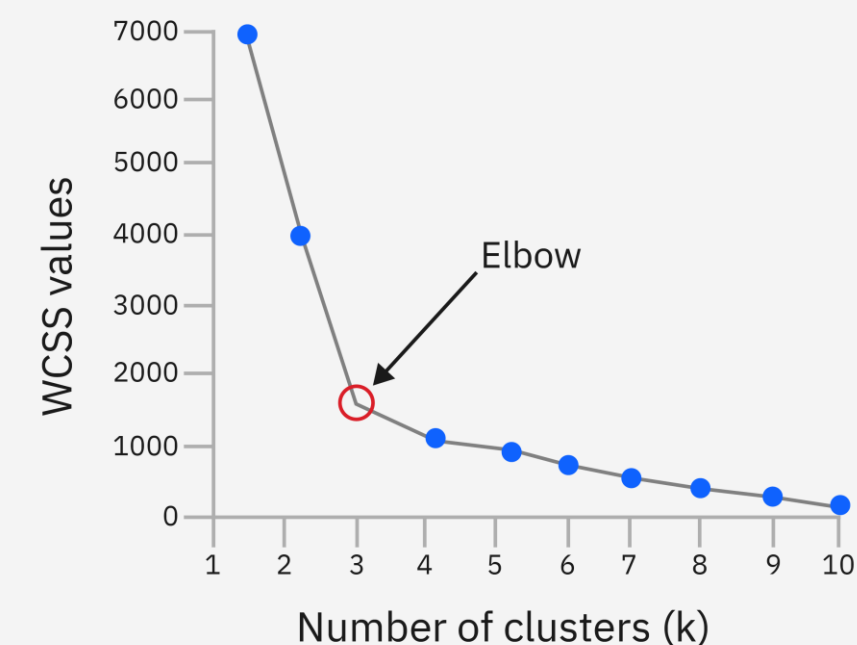
1. Choose k (number of clusters)
2. Assign k random centroids
3. Assign each observation to its nearest centroid
4. Recalculate centroids as cluster means
5. Repeat 3–4 until stable

- **Typology building** — replace subjective manual classification with a data-driven partition
- **Survey analysis** — identify latent respondent profiles
- **Policy targeting** — segment populations

How k-means groups the data



Choosing k — the elbow method



SECTION 07

Limitations of ML

Limits and Ethical Considerations

Conceptual limits

Correlation $\neq$ causation

ML finds patterns, not mechanisms.

Garbage in, garbage out

Model quality depends on data quality.

The black box problem

Complex models are hard to explain to reviewers, judges or policymakers.

Ethical risks

Encoded bias

If training data reflects historical inequality, the model reproduces it.

Differential accuracy

Models often perform worse for minority or underrepresented groups.

Feedback loops

Predictions inform decisions that generate new data, reinforcing the pattern.

A-level results: 'Huge mess' as exams appeal guidance withdrawn

16 August 2020



The exams regulator is reviewing its guidance on how to appeal against A-level and GCSE grades using mock exam results - hours after publishing it.

On Saturday Ofqual set out what constituted a "valid" mock exam for students appealing against A-level results in England.

The 2020 A-level grading algorithm was scrapped after it downgraded students from disadvantaged schools.

ML is a tool. The researcher's theoretical and ethical judgement remains indispensable.

SECTION 08

Today's Practical

Practical: 1994 US Census Income Data

- **48,842 adults** from the 1994 US Census
- Task: predict income $> \$50K$ or $\leq \$50K$
- Features: age, education, occupation, marital status, sex, race, hours/week, country of origin
- Relevant to labour market inequality, the gender pay gap and socioeconomic stratification

What we build

1. Decision tree classifier (with depth tuning)
2. Random forest classifier
3. Gradient boosting classifier
4. Extension: k-means clustering

How to access

- github.com/BHannan01/introduction-to-machine-learning →

Open in Colab

- No installation needed — Google Colab runs in the browser
- Data downloads automatically
- All code is heavily commented; reflection questions appear throughout

Questions?

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